

COMP1004-DBI

Databases and Interfaces

Module conveners:

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Learning Objectives

- How ER diagrams map to relational databases
- Normalisation
- Functional dependencies and their properties
- The process of normalisation (turning un-normalised data to 1NF, 1NF to 2NF, 2NF to 3NF)

Transforming ER diagram into relations

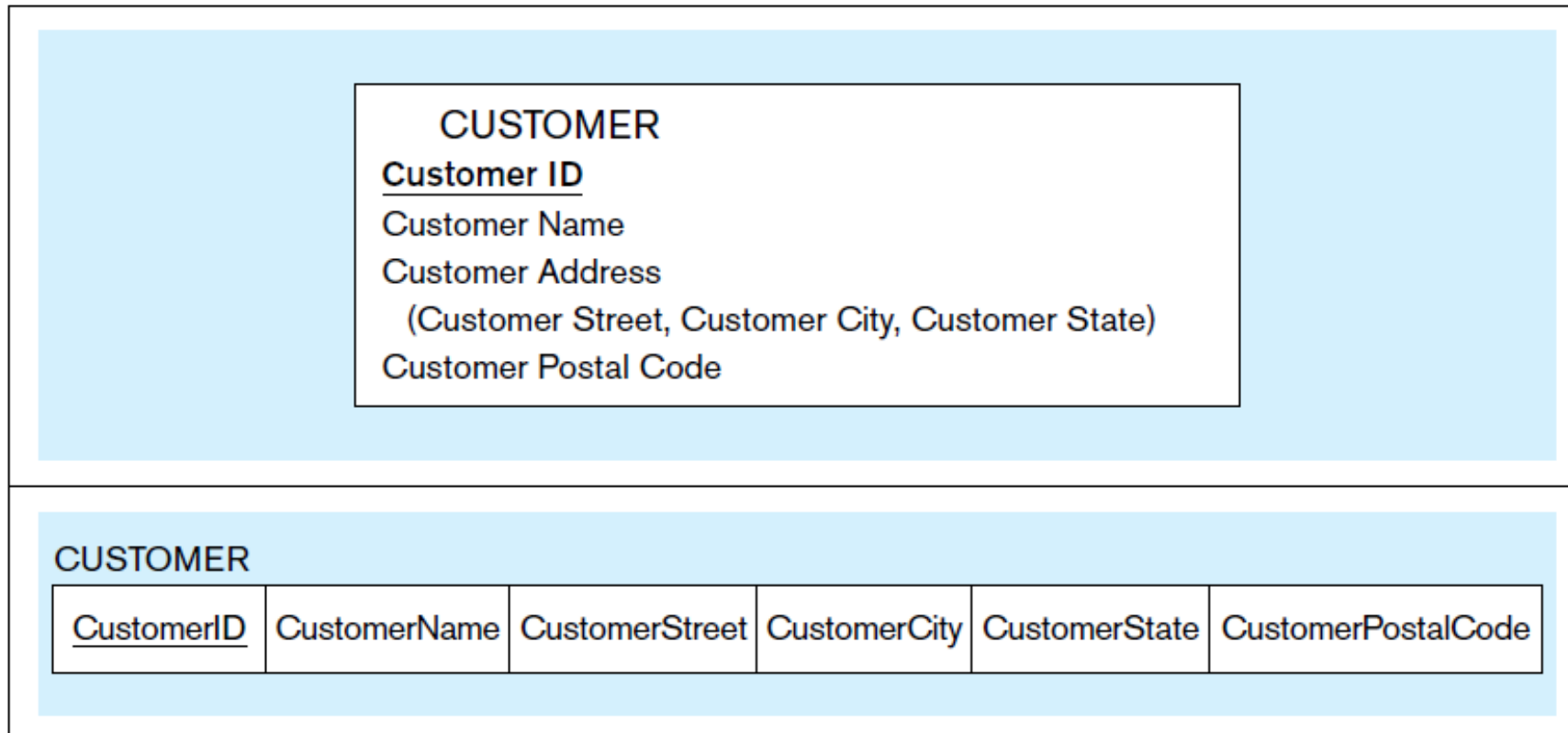
Transforming ER diagram into relations

- Transforming ER diagrams into relations is a straightforward process
- **Steps:**
- Step 1: Map Regular Entities
 - Each regular entity in an E-R diagram is transformed into a relation
 - The name given to the relation is generally the same as the entity
 - Each simple attribute of the entity becomes an attribute of the relation.
 - The identifier of the entity becomes the primary key of the corresponding relation.

Transforming ER diagram into relations

- Composite attributes

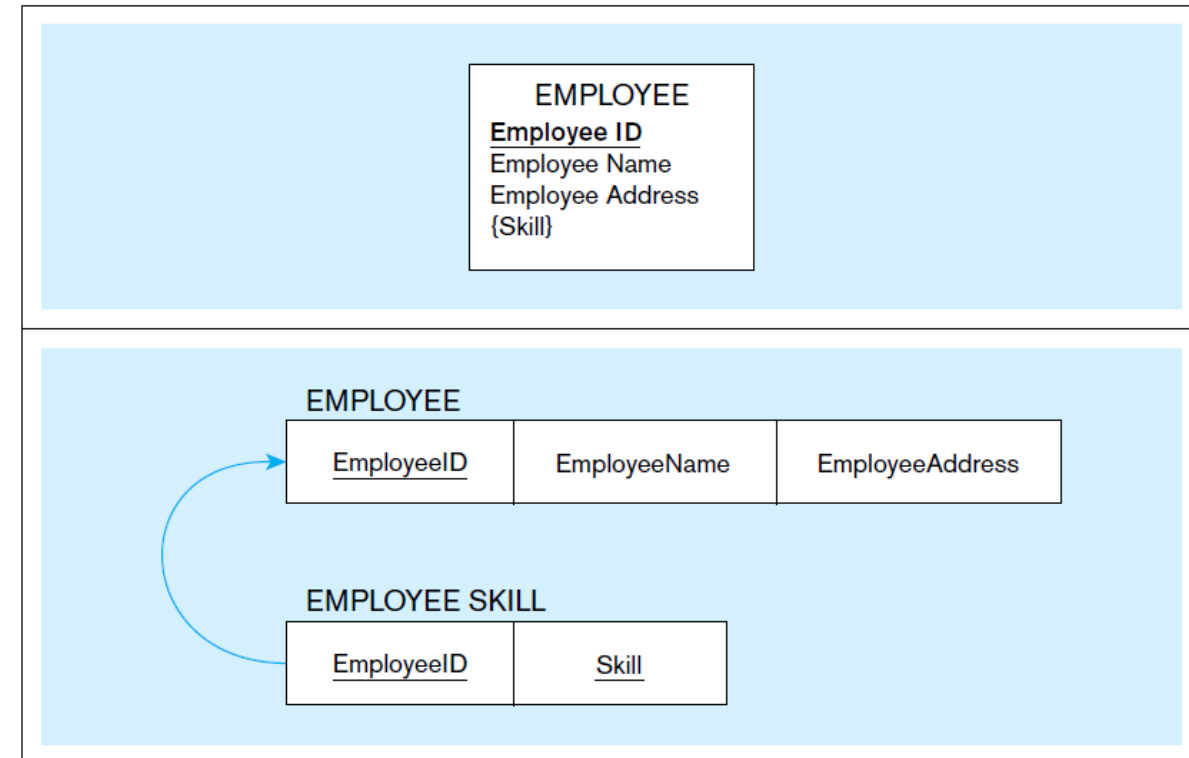
- When a regular entity has a composite attribute, only the simple components of the composite attribute are included in the new relation as its attributes.



Transforming ER diagram into relations

- Multivalued attributes

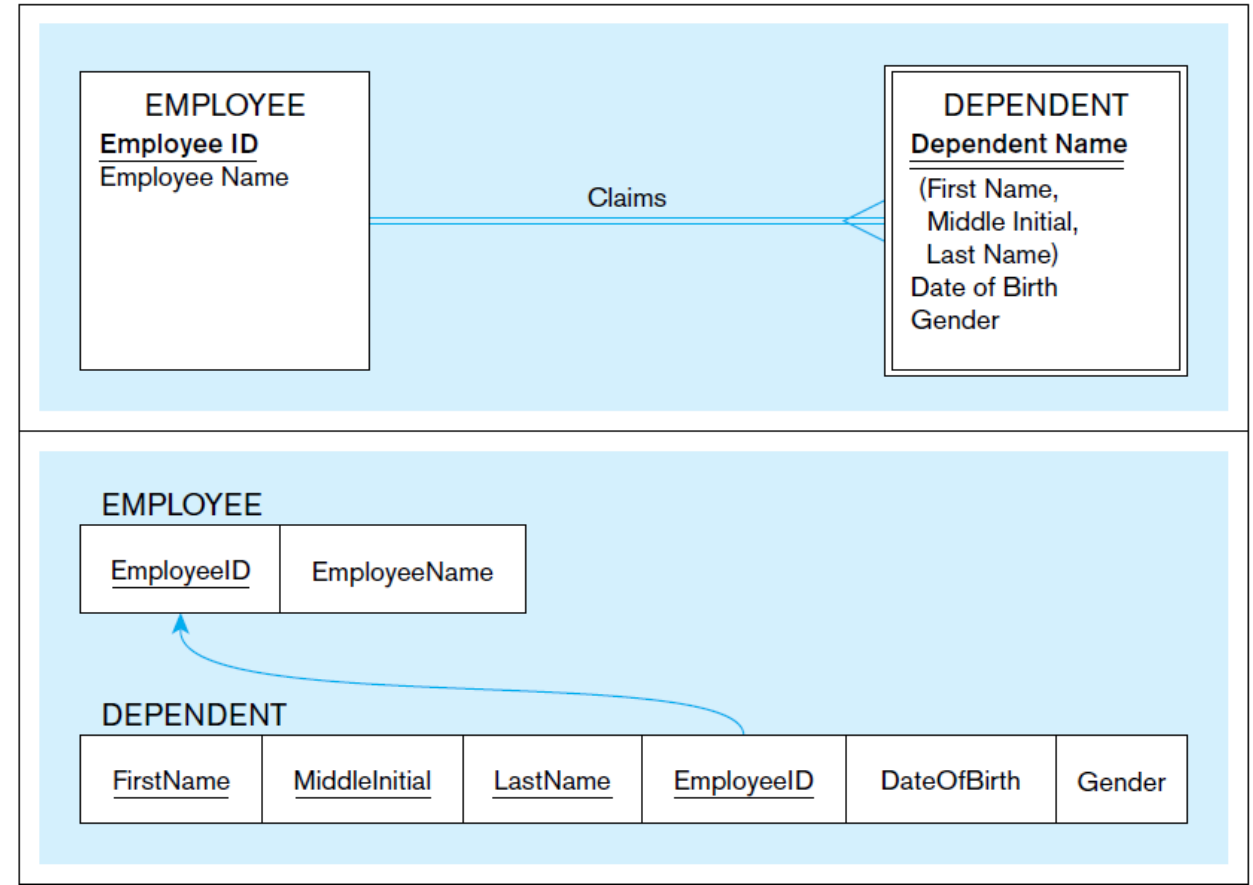
- When the regular entity contains a multivalued attribute, two new relations (rather than one) are created.
- The first relation contains all of the attributes of the entity type except the multivalued attribute.
- The second relation contains two attributes
 - 1st Primary key from the first relation
 - 2nd Multivalued attribute name



Transforming ER diagram into relations

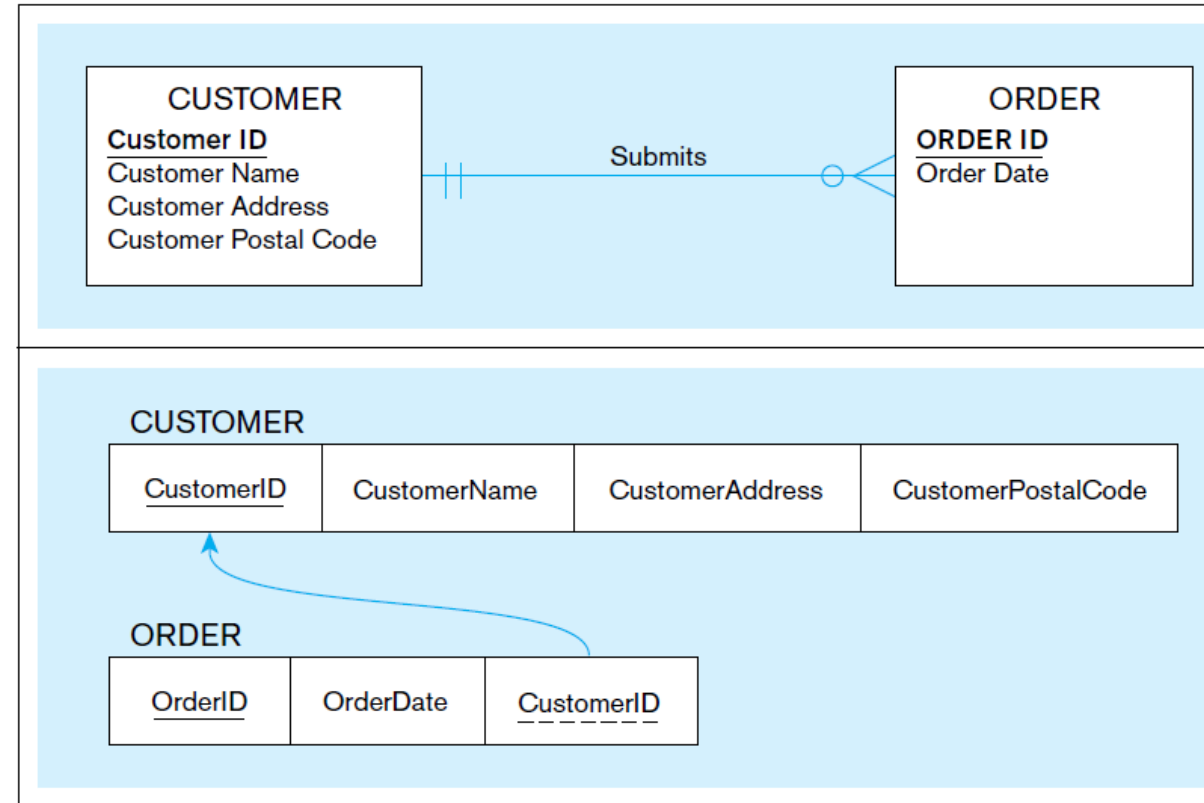
- Map Weak Entities

- For each weak entity, create a new relation
- Include all of the simple attributes as attributes of this relation.
- Include the primary key of the identifying relation as a foreign key attribute in this new relation



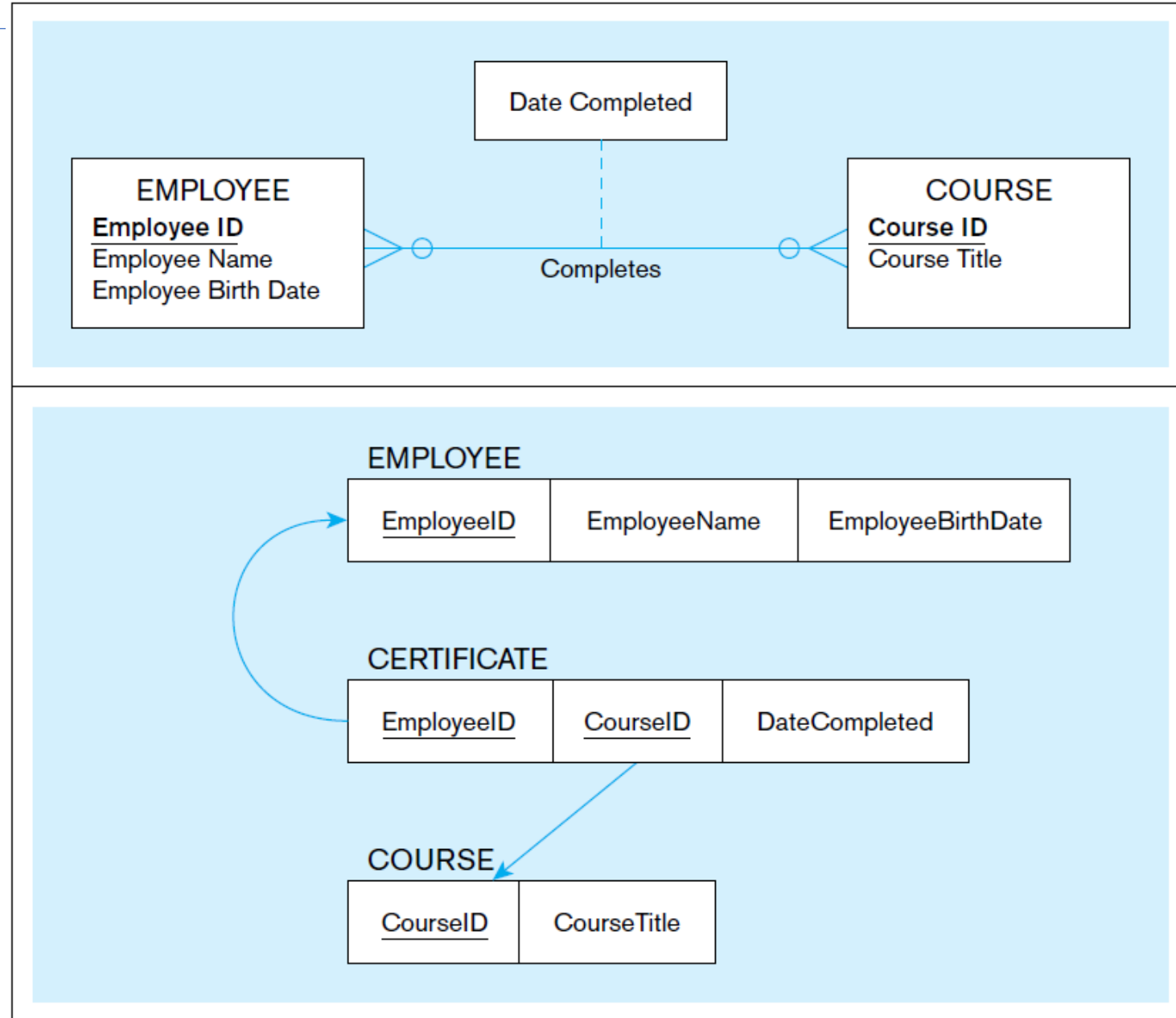
Transforming ER diagram into relations

- Map Binary Relationships
 - For each binary 1:M relationship, create a relation for each of the two entities participating in the relationship
 - Include the primary key attribute of the entity on the one-side of the relationship as a foreign key in the relation that is on the many-side of the relationship
 - HINT: (**The primary key migrates to the many side**)



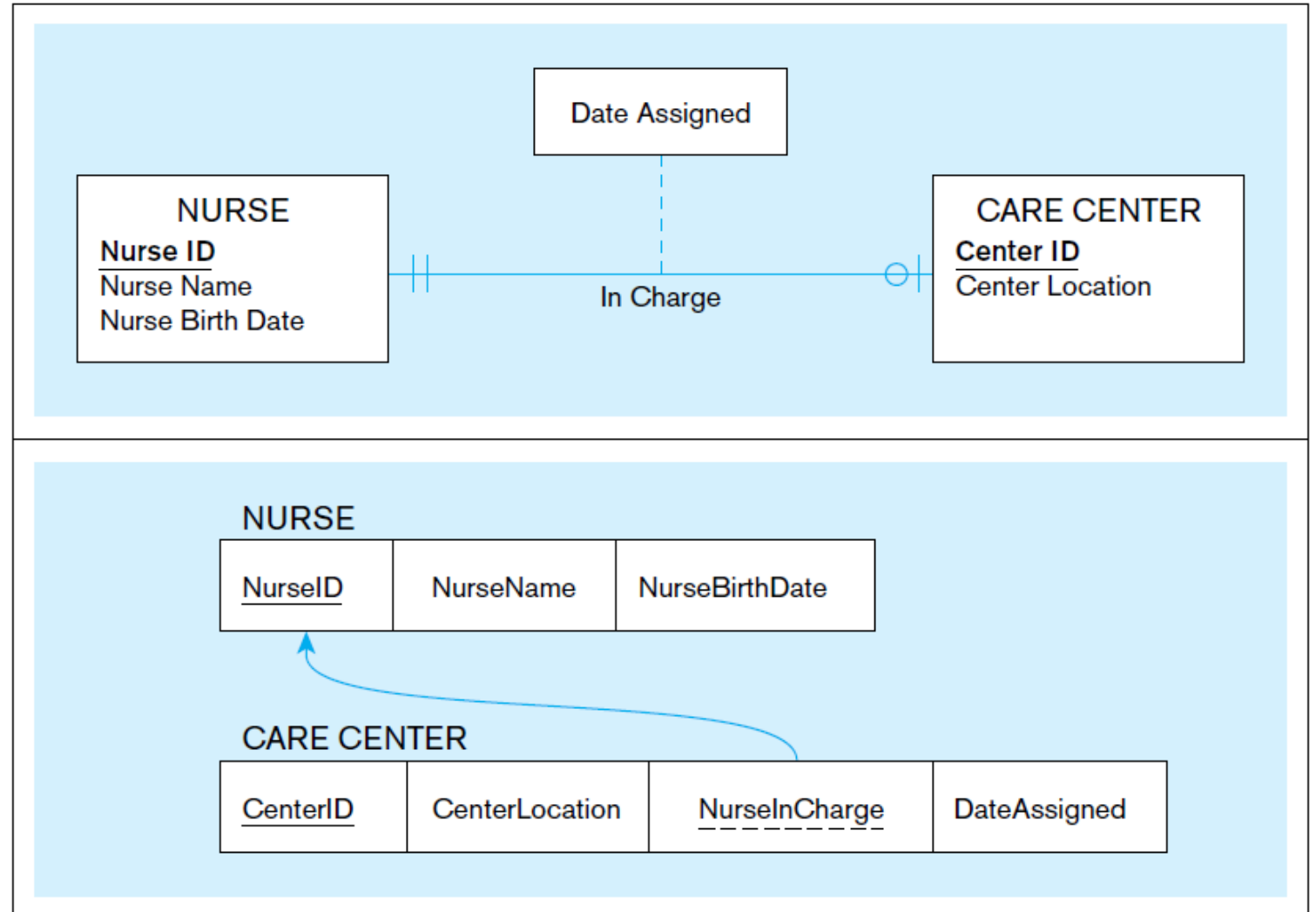
Transforming ER diagram into relations

- Map Binary Many-to-Many Relationships
 - Suppose that there is a binary many-to-many (M:N) relationship between two entities, A and B.
 - For such a relationship, create a new relation, C (associative entity)
 - The primary keys of A and B will act as foreign keys in C



Transforming ER diagram into relations

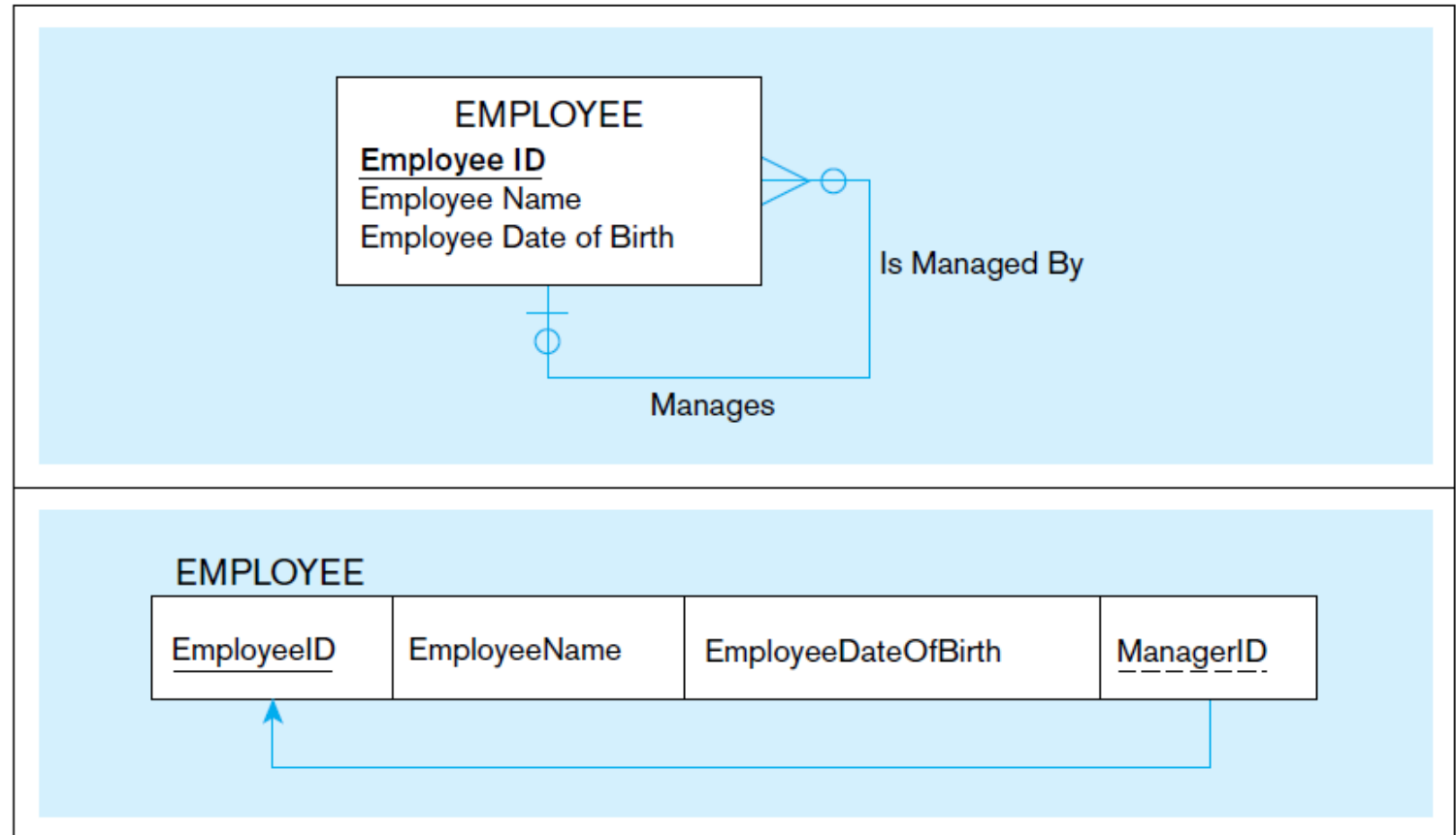
- Map Binary One-to-One Relationships
 - First, two relations are created, one for each of the participating entities
 - The primary key of one of the relations is included as a foreign key in the other relation.



Transforming ER diagram into relations

- Map Unary Relationships

- The entity in the unary relationship is mapped to a relation
- A foreign key attribute is added to the same relation; this attribute references the primary key values in the same relation



Normalisation

Normalisation

- An essential part of relational database theory and practice
- The process of decomposing relations with anomalies to produce smaller, well-structured relations.
- **Normalisation is a set of rules to reorganise relations to reduce redundancy and improve integrity**
- The normalisation process 'shakes out' these redundancies
- Principle:
 - Each table should have a single 'purpose'



Purpose of normalisation

- To identify a set of relations that meet the data needs of an organisation.
- A well-normalised database has the following characteristics:
 - It contains only the necessary attributes required to store data
 - Closely related attributes are stored together in the same table
 - Data redundancy is minimised, so each piece of data is stored only once

Why normalise?

- Data **anomalies**
- Anomalies are caused when there is too much **redundancy**
- **Inconsistencies** may arise when a particular record is stored in multiple locations and not all of the copies are (e.g.) updated
- Operations like updating, inserting, deleting all potentially introduce these inconsistencies
- Three types of anomalies:
 1. Insert Anomalies
 2. Delete Anomalies
 3. Update Anomalies

Data Redundancy and Anomalies

- StaffBranch (staffNo, sName, position, salary, branchNo, bAddress)
 - In the StaffBranch relation there is redundant data;
 - The details of a branch are repeated for every member of staff located at that branch.
 - Relations that have redundant data may have problems called anomalies
 - Insertion Anomalies
 - Deletion Anomalies
 - Modification Anomalies

StaffBranch

staffNo	sName	position	salary	branchNo	bAddress
SL21	John White	Manager	30000	B005	22 Deer Rd, London
SG37	Ann Beech	Assistant	12000	B003	163 Main St, Glasgow
SG14	David Ford	Supervisor	18000	B003	163 Main St, Glasgow
SA9	Mary Howe	Assistant	9000	B007	16 Argyll St, Aberdeen
SG5	Susan Brand	Manager	24000	B003	163 Main St, Glasgow
SL41	Julie Lee	Assistant	9000	B005	22 Deer Rd, London

Data Redundancy and Anomalies

- Insertion Anomalies

- To insert new member details, branch details must be included
- Every time branch details entered must be correct else there will be inconsistency
- Cannot insert new branch until it has a member
- If we insert a new branch data without member, we need to put null and staffno cannot be null (Entity integrity constraint will be violated)

StaffBranch

staffNo	sName	position	salary	branchNo	bAddress
SL21	John White	Manager	30000	B005	22 Deer Rd, London
SG37	Ann Beech	Assistant	12000	B003	163 Main St, Glasgow
SG14	David Ford	Supervisor	18000	B003	163 Main St, Glasgow
SA9	Mary Howe	Assistant	9000	B007	16 Argyll St, Aberdeen
SG5	Susan Brand	Manager	24000	B003	163 Main St, Glasgow
SL41	Julie Lee	Assistant	9000	B005	22 Deer Rd, London

Data Redundancy and Anomalies

- Deletion Anomalies

- If we delete the tuple for staff number SA9 (Mary Howe), branch B007 details will be lost

StaffBranch

staffNo	sName	position	salary	branchNo	bAddress
SL21	John White	Manager	30000	B005	22 Deer Rd, London
SG37	Ann Beech	Assistant	12000	B003	163 Main St, Glasgow
SG14	David Ford	Supervisor	18000	B003	163 Main St, Glasgow
SA9	Mary Howe	Assistant	9000	B007	16 Argyll St, Aberdeen
SG5	Susan Brand	Manager	24000	B003	163 Main St, Glasgow
SL41	Julie Lee	Assistant	9000	B005	22 Deer Rd, London

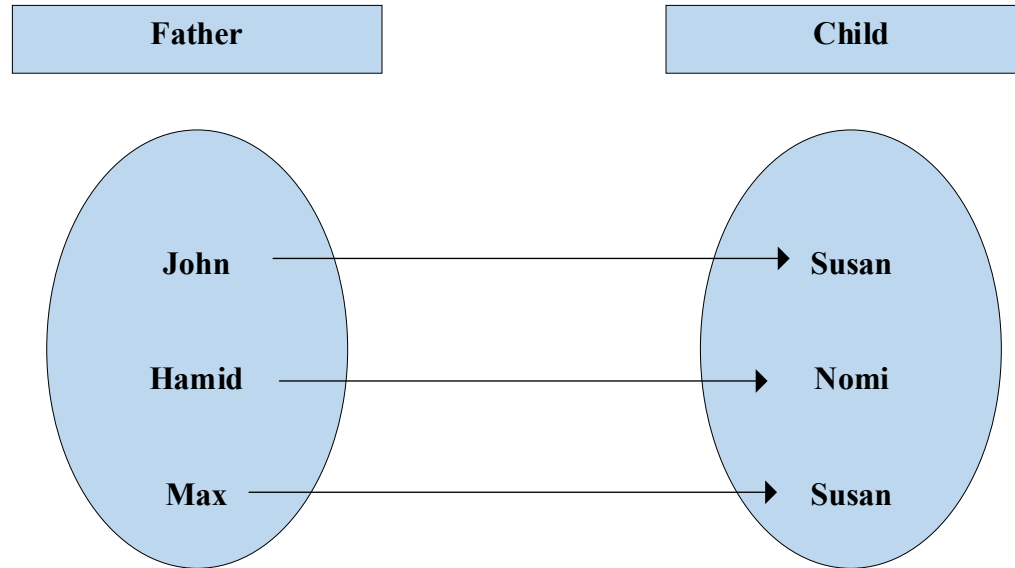
- Modification Anomalies

- If the address for branch number B003 is changed, it need to be changed at three locations
- If any mistake is made while updating any one record, there will be inconsistent data

StaffBranch

staffNo	sName	position	salary	branchNo	bAddress
SL21	John White	Manager	30000	B005	22 Deer Rd, London
SG37	Ann Beech	Assistant	12000	B003	163 Main St, Glasgow
SG14	David Ford	Supervisor	18000	B003	163 Main St, Glasgow
SA9	Mary Howe	Assistant	9000	B007	16 Argyll St, Aberdeen
SG5	Susan Brand	Manager	24000	B003	163 Main St, Glasgow
SL41	Julie Lee	Assistant	9000	B005	22 Deer Rd, London

Functional Dependencies- Example



Functional Dependencies

StudentID	StudentName	Major
101	Alex	Computer science
102	Susan	Math
103	Nomi	Physics
104	Fabiha	Computer science

- $\text{StudentID} \rightarrow \text{StudentName}$. Does functional dependency exist?

- a) True
- b) False

Functional Dependencies

StudentID	StudentName	Major
101	Alex	Computer science
102	Susan	Math
103	Nomi	Physics
104	Fabiha	Computer science

- Major $\rightarrow$ StudentID. Does functional dependency exist?
 - a) True
 - b) False

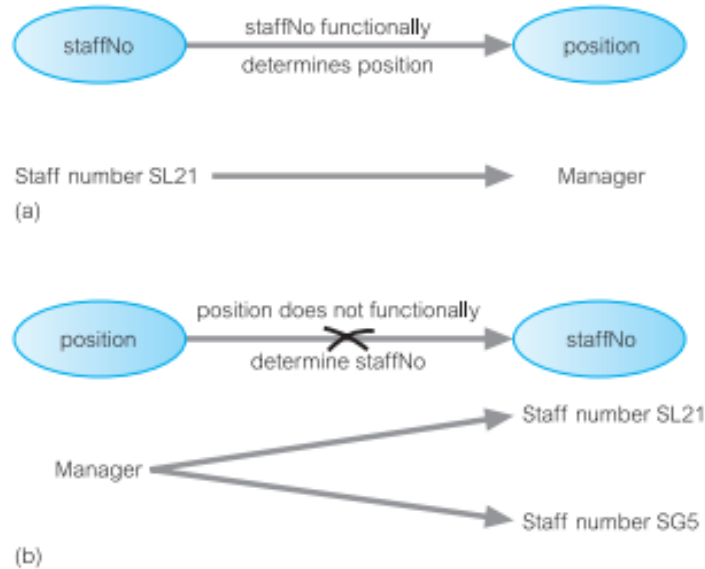
Functional Dependencies

- Functional dependency:
 - A constraint between two attributes in which the value of one attribute is determined by the value of another attribute.
 - For example, if **A** and **B** are attributes of relation **R**, B is functionally dependent on A (denoted $A \twoheadrightarrow B$, or $A \rightarrow B$), if each value of A is associated with exactly one value of B.
 - A is known as **determinant**
 - For a value of A there will be one value of B, viceversa is not true



Functional Dependencies

- Example 1



Staff Branch

staffNo	sName	position	salary	branchNo	bAddress
SL21	John White	Manager	30000	B005	22 Deer Rd, London
SG37	Ann Beech	Assistant	12000	B003	163 Main St, Glasgow
SG14	David Ford	Supervisor	18000	B003	163 Main St, Glasgow
SA9	Mary Howe	Assistant	9000	B007	16 Argyll St, Aberdeen
SG5	Susan Brand	Manager	24000	B003	163 Main St, Glasgow
SL41	Julie Lee	Assistant	9000	B005	22 Deer Rd, London

- Example 2

- **staffNo** ® **sName** or **staffNo** → **sName**
- **sName** ® **staffNo** or **sName** → **staffNo** (true only for the given dataset)

Full functional dependency

- A functional dependency $A \rightarrow B$ is a full functional dependency if removal of any attribute from A results in the dependency no longer existing

$(\text{StudentID}, \text{CourseID}) \rightarrow \text{Grade}$

- A functional dependency $A \rightarrow B$ is a partial dependency if there is some attribute that can be removed from A and yet the dependency still holds.

$(\text{StudentID}, \text{CourseID}) \rightarrow \text{StudentName}$

$(\text{StaffNo}, \text{sName}) \rightarrow \text{branchNo}$

StaffBranch

staffNo	sName	position	salary	branchNo	bAddress
SL21	John White	Manager	30000	B005	22 Deer Rd, London
SG37	Ann Beech	Assistant	12000	B003	163 Main St, Glasgow
SG14	David Ford	Supervisor	18000	B003	163 Main St, Glasgow
SA9	Mary Howe	Assistant	9000	B007	16 Argyll St, Aberdeen
SG5	Susan Brand	Manager	24000	B003	163 Main St, Glasgow
SL41	Julie Lee	Assistant	9000	B005	22 Deer Rd, London

Transitive dependency

- A functional dependency between the primary key and one or more nonkey attributes that are dependent on the primary key via another nonkey attribute.
- Example
 - $\text{staffNo} \rightarrow \text{sName}, \text{position}, \text{salary}, \text{branchNo}, \text{bAddress}$
 - $\text{branchNo} \rightarrow \text{bAddress}$
 - The transitive dependency $\text{branchNo} \rightarrow \text{bAddress}$ exists on staffNo via branchNo
And
 - neither branchNo nor bAddress functionally determines staffNo

StaffBranch

staffNo	sName	position	salary	branchNo	bAddress
SL21	John White	Manager	30000	B005	22 Deer Rd, London
SG37	Ann Beech	Assistant	12000	B003	163 Main St, Glasgow
SG14	David Ford	Supervisor	18000	B003	163 Main St, Glasgow
SA9	Mary Howe	Assistant	9000	B007	16 Argyll St, Aberdeen
SG5	Susan Brand	Manager	24000	B003	163 Main St, Glasgow
SL41	Julie Lee	Assistant	9000	B005	22 Deer Rd, London

Dependencies

Convenerid, Module → Convenerage

CONVENER

<u>Convenerid</u>	<u>Module</u>	<u>Convenerage</u>
Psz001	COMP1001	30
Psz001	COMP1002	30
Psz002	COMP1003	35
Psz004	COMP1004	38
Psz004	COMP1005	38

Which dependency exists?

- a) Full functional dependency
- b) Partial dependency

Dependencies

Convenerid → Convenername, Convenerpostcode, Convenercity

CONVENER

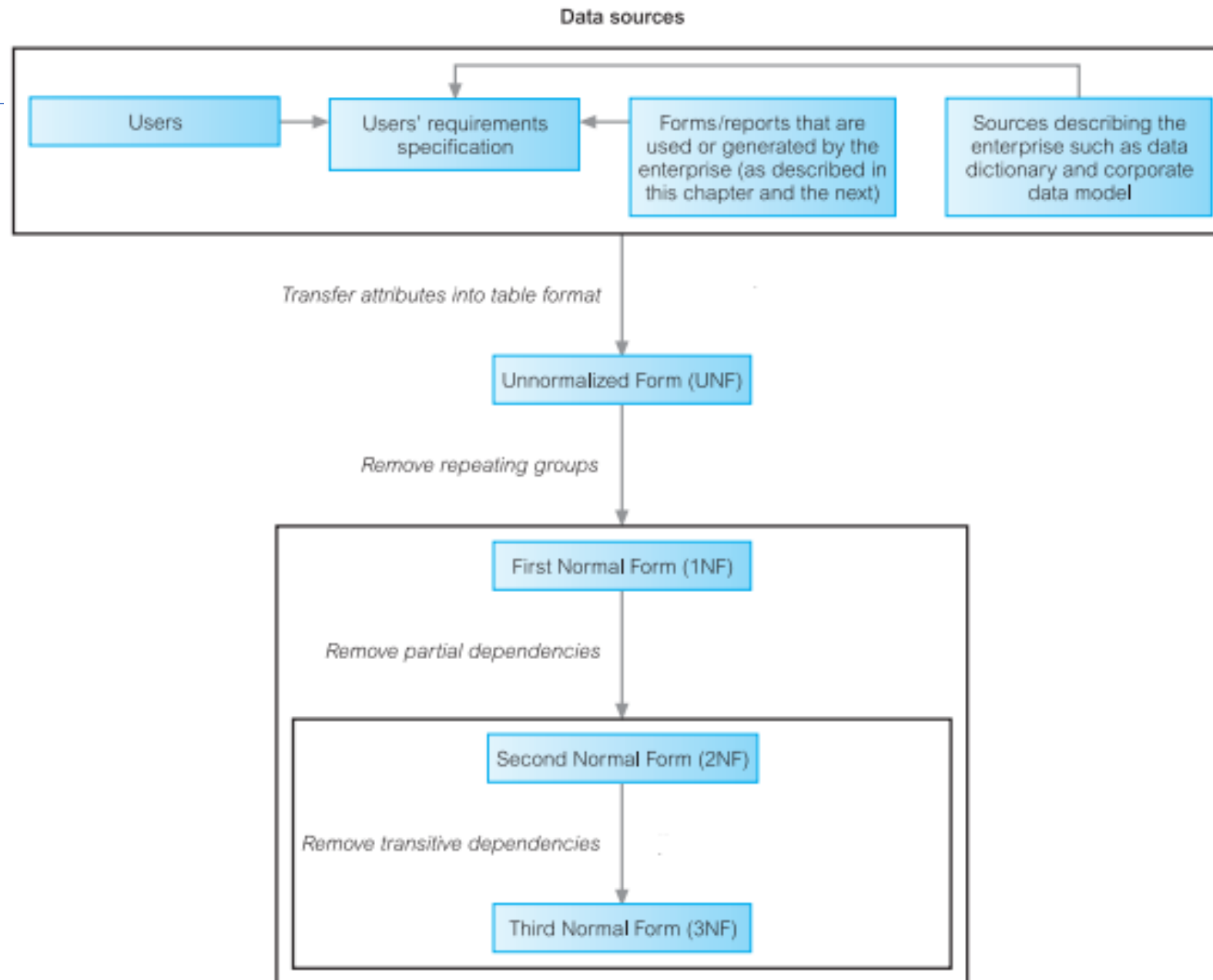
<u>Convenerid</u>	Convenername	Convenerpostcode	Convenercity
Psz001	Jeremie	NG8 6TY	Nottingham
Psz002	Susan	AB10 0UY	Aberdeen
Psz003	Warren	B11 7BN	Birmingham
Psz004	Liz	BS16 4RT	Bristol
Psz005	Warwara	CF3 9GH	Cardiff

Which dependency exists?

- a) Full functional dependency
- b) Partial dependency
- c) Transitive dependency

Steps in Normalisation

- First normal form
 - Any multivalued attributes (also called repeating groups) have been removed, so there is a single value at the intersection of each row and column of the table
- Second normal form
 - Any partial functional dependencies have been removed (i.e., nonkey attributes are identified by the whole primary key).
- Third normal form
 - Any transitive dependencies have been removed (i.e., nonkey attributes are identified by only the primary key).



Conclusion: Characteristics of normalisation

- **Lossless**
 - Data should not be lost or created when splitting relations up
- **Dependency preservation**
 - It is desirable that FDs are preserved when splitting relations up
- Normalisation to 3NF is always lossless and dependency preserving

Un-normalise form

ClientRental

clientNo	cName	propertyNo	pAddress	rentStart	rentFinish	rent	ownerNo	oName
CR76	John Kay	PG4	6 Lawrence St, Glasgow	1-Jul-12	31-Aug-13	350	CO40	Tina Murphy
		PG16	5 Novar Dr, Glasgow	1-Sep-13	1-Sep-14	50	CO93	Tony Shaw
CR56	Aline Stewart	PG4	6 Lawrence St, Glasgow	1-Sep-11	10-June-12	350	CO40	Tina Murphy
		PG36	2 Manor Rd, Glasgow	10-Oct-12	1-Dec-13	375	CO93	Tony Shaw
		PG16	5 Novar Dr, Glasgow	1-Nov-14	10-Aug-15	450	CO93	Tony Shaw

Normalisation (example)

- 1st Normal Form
- There are two common approaches to removing repeating groups from unnormalised tables
 - By entering appropriate data in the empty columns of rows containing the repeating data. 
 - By placing the repeating data, along with a copy of the original key attribute(s), in a separate relation 

ClientRental

clientNo	cName	propertyNo	pAddress	rentStart	rentFinish	rent	ownerNo	oName
CR76	John Kay	PG4	6 Lawrence St, Glasgow	1-Jul-12	31-Aug-13	350	CO40	Tina Murphy
		PG16	5 Novar Dr, Glasgow	1-Sep-13	1-Sep-14	50	CO93	Tony Shaw
CR56	Aline Stewart	PG4	6 Lawrence St, Glasgow	1-Sep-11	10-June-12	350	CO40	Tina Murphy
		PG36	2 Manor Rd, Glasgow	10-Oct-12	1-Dec-13	375	CO93	Tony Shaw
		PG16	5 Novar Dr, Glasgow	1-Nov-14	10-Aug-15	450	CO93	Tony Shaw

ClientRental

clientNo	propertyNo	cName	pAddress	rentStart	rentFinish	rent	ownerNo	oName
CR76	PG4	John Kay	6 Lawrence St, Glasgow	1-Jul-12	31-Aug-13	350	CO40	Tina Murphy
CR76	PG16	John Kay	5 Novar Dr, Glasgow	1-Sep-13	1-Sep-14	450	CO93	Tony Shaw
CR56	PG4	Aline Stewart	6 Lawrence St, Glasgow	1-Sep-11	10-Jun-12	350	CO40	Tina Murphy
CR56	PG36	Aline Stewart	2 Manor Rd, Glasgow	10-Oct-12	1-Dec-13	375	CO93	Tony Shaw
CR56	PG16	Aline Stewart	5 Novar Dr, Glasgow	1-Nov-14	10-Aug-15	450	CO93	Tony Shaw

Client

clientNo	cName
CR76	John Kay
CR56	Aline Stewart

PropertyRentalOwner

clientNo	propertyNo	pAddress	rentStart	rentFinish	rent	ownerNo	oName
CR76	PG4	6 Lawrence St, Glasgow	1-Jul-12	31-Aug-13	350	CO40	Tina Murphy
CR76	PG16	5 Novar Dr, Glasgow	1-Sep-13	1-Sep-14	450	CO93	Tony Shaw
CR56	PG4	6 Lawrence St, Glasgow	1-Sep-11	10-Jun-12	350	CO40	Tina Murphy
CR56	PG36	2 Manor Rd, Glasgow	10-Oct-12	1-Dec-13	375	CO93	Tony Shaw
CR56	PG16	5 Novar Dr, Glasgow	1-Nov-14	10-Aug-15	450	CO93	Tony Shaw

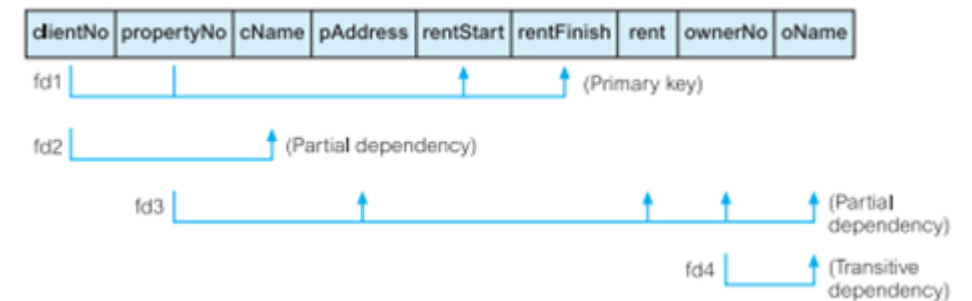
Normalisation

- Functional dependencies of the ClientRental relation.
 - fd1** clientNo, propertyNo → rentStart, rentFinish (Primary key)
 - fd2** clientNo → cName (Partial dependency)
 - fd3** propertyNo → pAddress, rent, ownerNo, oName (Partial dependency)
 - fd4** ownerNo → oName (Transitive dependency)

ClientRental

clientNo	propertyNo	cName	pAddress	rentStart	rentFinish	rent	ownerNo	oName
CR76	PG4	John Kay	6 Lawrence St, Glasgow	1-Jul-12	31-Aug-13	350	CO40	Tina Murphy
CR76	PG16	John Kay	5 Novar Dr, Glasgow	1-Sep-13	1-Sep-14	450	CO93	Tony Shaw
CR56	PG4	Aline Stewart	6 Lawrence St, Glasgow	1-Sep-11	10-Jun-12	350	CO40	Tina Murphy
CR56	PG36	Aline Stewart	2 Manor Rd, Glasgow	10-Oct-12	1-Dec-13	375	CO93	Tony Shaw
CR56	PG16	Aline Stewart	5 Novar Dr, Glasgow	1-Nov-14	10-Aug-15	450	CO93	Tony Shaw

ClientRental



Normalisation (example)

- ClientRental (clientNo, propertyNo, cName, pAddress, rentStart, rentFinish, rent, ownerNo, oName)
- Partial dependency
 - clientNo $\rightarrow$ cName
 - propertyNo $\rightarrow$ pAddress, rent, ownerNo, oName
- Second normal form (2NF)
 - Relation is in 1st normal form
 - Any partial functional dependencies have been removed

ClientRental

clientNo	propertyNo	cName	pAddress	rentStart	rentFinish	rent	ownerNo	oName
CR76	PG4	John Kay	6 Lawrence St, Glasgow	1-Jul-12	31-Aug-13	350	CO40	Tina Murphy
CR76	PG16	John Kay	5 Novar Dr, Glasgow	1-Sep-13	1-Sep-14	450	CO93	Tony Shaw
CR56	PG4	Aline Stewart	6 Lawrence St, Glasgow	1-Sep-11	10-Jun-12	350	CO40	Tina Murphy
CR56	PG36	Aline Stewart	2 Manor Rd, Glasgow	10-Oct-12	1-Dec-13	375	CO93	Tony Shaw
CR56	PG16	Aline Stewart	5 Novar Dr, Glasgow	1-Nov-14	10-Aug-15	450	CO93	Tony Shaw

ClientRental

clientNo	propertyNo	cName	pAddress	rentStart	rentFinish	rent	ownerNo	oName
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Client

clientNo	cName
CR76	John Kay
CR56	Aline Stewart

PropertyOwner

propertyNo	pAddress	rent	ownerNo	oName
PG4	6 Lawrence St, Glasgow	350	CO40	Tina Murphy
PG16	5 Novar Dr, Glasgow	450	CO93	Tony Shaw
PG36	2 Manor Rd, Glasgow	375	CO93	Tony Shaw

Rental

clientNo	propertyNo	rentStart	rentFinish
CR76	PG4	1-Jul-12	31-Aug-13
CR76	PG16	1-Sep-13	1-Sep-14
CR56	PG4	1-Sep-11	10-Jun-12
CR56	PG36	10-Oct-12	1-Dec-13
CR56	PG16	1-Nov-14	10-Aug-15

Second normal form relations derived from the ClientRental relation.

Normalisation (example)

- Third Normal Form (3NF)
 - A relation is in 2nd normal form
 - Any transitive dependencies have been removed (i.e., nonkey attributes are identified by only the primary key).

ClientRental

clientNo	propertyNo	cName	pAddress	rentStart	rentFinish	rent	ownerNo	oName
CR76	PG4	John Kay	6 Lawrence St, Glasgow	1-Jul-12	31-Aug-13	350	CO40	Tina Murphy
CR76	PG16	John Kay	5 Novar Dr, Glasgow	1-Sep-13	1-Sep-14	450	CO93	Tony Shaw
CR56	PG4	Aline Stewart	6 Lawrence St, Glasgow	1-Sep-11	10-Jun-12	350	CO40	Tina Murphy
CR56	PG36	Aline Stewart	2 Manor Rd, Glasgow	10-Oct-12	1-Dec-13	375	CO93	Tony Shaw
CR56	PG16	Aline Stewart	5 Novar Dr, Glasgow	1-Nov-14	10-Aug-15	450	CO93	Tony Shaw

ClientRental

clientNo	propertyNo	cName	pAddress	rentStart	rentFinish	rent	ownerNo	oName
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fd4  (Transitive dependency)

PropertyForRent

propertyNo	pAddress	rent	ownerNo
PG4	6 Lawrence St, Glasgow	350	CO40
PG16	5 Novar Dr, Glasgow	450	CO93
PG36	2 Manor Rd, Glasgow	375	CO93

Owner

ownerNo	oName
CO40	Tina Murphy
CO93	Tony Shaw

Third normal form relations derived from the PropertyOwner relation

Normalisation (example)

- The decomposition of the ClientRental 1NF relation into 3NF relations.
 - Client (clientNo, cName)
 - Rental (clientNo, propertyNo, rentStart, rentFinish)
 - PropertyForRent (propertyNo, pAddress, rent, ownerNo)
 - Owner (ownerNo, oName)



Client

clientNo	cName
CR76	John Kay
CR56	Aline Stewart

PropertyForRent

propertyNo	pAddress	rent	ownerNo
PG4	6 Lawrence St, Glasgow	350	CO40
PG16	5 Novar Dr, Glasgow	450	CO93
PG36	2 Manor Rd, Glasgow	375	CO93

Owner

ownerNo	oName
CO40	Tina Murphy
CO93	Tony Shaw

Rental

clientNo	propertyNo	rentStart	rentFinish
CR76	PG4	1-Jul-12	31-Aug-13
CR76	PG16	1-Sep-13	1-Sep-14
CR56	PG4	1-Sep-11	10-Jun-12
CR56	PG36	10-Oct-12	1-Dec-13
CR56	PG16	1-Nov-14	10-Aug-15

Example

Shopping Invoice				
Invoice Number:IN1 Customer No:001 Customer Address:NG6 6TY, Nottingham Invoice Date:01/01/ 2024				
ItemCode	ItemDesc	ItemPrice(ItPr)	Quantity(Q)	Price(ItPr*Q)
B001	Bread	1.50	1	1.50
F001	Apple	2	2	4
M001	Milk	1.50	1	1.50
Total(£):7				

Shopping Invoice				
Invoice Number:IN2 Customer No:002 Customer Address:NG8 5HT, Nottingham Invoice Date:02/01/ 2024				
ItemCode	ItemDesc	ItemPrice(ItPr)	Quantity(Q)	Price(ItPr*Q)
B001	Bread	1.5	2	3
Total(£):3				

Shopping Invoice				
Invoice Number:IN3 Customer No:004 Customer Address:NG1, 6TT, Nottingham Invoice Date:02/01/ 2024				
ItemCode	ItemDesc	ItemPrice(ItPr)	Quantity(Q)	Price(ItPr*Q)
V004	Potato	1	2	2
B002	Bread	2	1	2
F001	Apple	2	2	4
Total(£):8				

Non-Normal form

InvoiceNo	CustNo	CustAddress	InvoiceDate	ItemCode	ItemDesc	ItemPrice(ItPr)	Quantity(Q)	Price(ItPr*Q)	InoviceTotal
IN1	001	NG6 6TY, Nottingham	01/01/2024	B001	Bread	1.50	1	1.50	7
				F001	Apple	2	2	4	
				M001	Milk	1.50	1	1.50	
IN2	002	NG8 5HT, Nottingham	02/01/2024	B001	Bread	1.50	2	3	3
IN3	004	NG1, 6TT, Nottingham	02/01/2024	V004	Potato	1	2	2	8
				B002	Bread	2	1	2	8
				F001	Apple	2	2	4	8

1st Normal form

<u>InvoiceNo</u>	CustNo	CustAddress	InvoiceDate	<u>ItemCode</u>	ItemDesc	ItemPrice(ItPr)	Quantity(Q)	Price(ItPr*Q)	InoviceTotal
IN1	001	NG6 6TY, Nottingham	01/01/2024	B001	Bread	1.50	1	1.50	7
IN1	001	NG6 6TY, Nottingham	01/01/2024	F001	Apple	2	2	4	7
IN1	001	NG6 6TY, Nottingham	01/01/2024	M001	Milk	1.50	1	1.50	7
IN2	002	NG8 5HT, Nottingham	02/01/2024	B001	Bread	1.50	2	3	3
IN3	004	NG1, 6TT, Nottingham	02/01/2024	V004	Potato	1	2	2	8
IN3	004	NG1, 6TT, Nottingham	02/01/2024	B002	Bread	2	1	2	8
IN3	004	NG1, 6TT, Nottingham	02/01/2024	F001	Apple	2	2	4	8

Primary key (InvoiceNo, ItemCode)

Dependencies

InvoiceNo → CustNo, CustAddress, InvoiceDate, InoviceTotal (partial Dependency)

ItemCode → ItemDesc, ItemPrice (Partial Dependency)

CustNo → CustAddress

2nd Normal Form

INVOICE

<u>InvoiceNo</u>	CustNo	CustAddress	InvoiceDate	InoviceTotal
IN1	001	NG6 6TY, Nottingham	01/01/2024	7
IN2	002	NG8 5HT, Nottingham	02/01/2024	3
IN3	004	NG1, 6TT, Nottingham	02/01/2024	8

ITEM

<u>ItemCode</u>	ItemDesc	ItemPrice(ItPr)
B001	Bread	1.50
F001	Apple	2
M001	Milk	1.50
V004	Potato	1
B002	Bread	2

INVOICEDetail

<u>InvoiceNo</u>	<u>ItemCode</u>	Quantity(Q)	Price(ItPr*Q)
IN1	B001	1	1.50
IN1	F001	2	4
IN1	M001	1	1.50
IN2	B001	2	3
IN3	V004	2	2
IN3	B002	1	2
IN3	F001	2	4

3rd Normal Form

INVOICE

<u>InvoiceNo</u>	CustNo	InvoiceDate	InoviceTotal
IN1	001	01/01/2024	7
IN2	002	02/01/2024	3
IN3	004	02/01/2024	8

INVOICEDetail

<u>InvoiceNo</u>	<u>ItemCode</u>	Quantity(Q)	Price(ItPr*Q)
IN1	B001	1	1.50
IN1	F001	2	4
IN1	M001	1	1.50
IN2	B001	2	3
IN3	V004	2	2
IN3	B002	1	2
IN3	F001	2	4

CUSTOMER

<u>CustNo</u>	CustAddress
001	NG6 6TY, Nottingham
002	NG8 5HT, Nottingham
004	NG1, 6TT, Nottingham

ITEM

<u>ItemCode</u>	ItemDesc	ItemPrice(ItPr)
B001	Bread	1.50
F001	Apple	2
M001	Milk	1.50
V004	Potato	1
B002	Bread	2

Home Exercise (Transform to 3rd Normal form)

REPORT CODE	DATE	EXPORT DEST. CODE	EXPORT DEST. NAME	PRODUCT CODE	PRODUCT NAME	UNIT PRICE	QUANTITY
1101	3/5	12	THE KINGDOM OF MINANMI	101	MELON	800G	1,100
				102	STRAWBERRY	150G	300
1102	3/7	23	ALPHA EMPIRE	103	APPLE	120G	1,700
1103	3/8	25	THE KINGDOM OF RITOL	104	LEMON	200G	500
1104	3/10	12	THE KINGDOM OF MINANMI	101	MELON	800G	2,500
1105	3/12	25	THE KINGDOM OF RITOL	103	APPLE	120G	2,000
				104	LEMON	200G	700